1. Clinical Question & Core Hypothesis

Clinical Question

Why do TLS-like lymphoid aggregates not uniformly confer a survival advantage in breast cancer—i.e., what tumor-intrinsic factors convert histologic TLS presence from a beneficial structure into a functionally impotent or even permissive state? 2) Tier 1 paths used: I built on the CXCL pathway path (PTEN → PI3K → chemokine milieu) and the TIL cluster C4 / mTORC1 path (PTEN → PI3K/AKT → mTORC1 negatively correlates with TLS-associated TIL phenotypes). These link PTEN loss to altered chemokine-driven recruitment, endothelial/antigen-presentation changes, and a specific TIL phenotype tied to prognosis, providing a direct mechanistic route for TLS dysfunction. 3) Tier 2 support: PAK, YAP/TAZ, and TLR/NO paths strengthen the plausibility that PTEN-driven signaling remodels cytoskeleton, mechano-transduction, and innate inflammation—each capable of disrupting immune cell trafficking or APC function, thereby weakening TLS efficacy. 4) Endpoint choice: Overall survival (OS) was selected because TCGA-BRCA provides robust OS data and the question addresses a durable clinical consequence (not immediate therapy response). 5) Deliberately excluded: I excluded listing many individual chemokine genes or HLA alleles in the core hypothesis to keep the mechanistic claim sharp (max 2 anchors). Exploratory/co-expression modules (e.g., turquoise module) and cross-tumor context links (glioma/prostate-specific chains) were deprioritized as they are less directly mechanistic in breast cancer context?

Working Hypothesis

TLS-like lymphoid aggregates are prognostically beneficial in breast cancer when the tumor microenvironment supports effective recruitment and organization of B and T cells, efficient antigen presentation, and proper endothelial maturation. I hypothesize that PTEN loss with concomitant mTORC1 hyperactivation reprograms tumor and stromal compartments to suppress TLS function. Specifically, PTEN loss will be associated with reduced CXCL13 and related TLS-supporting chemokines in TLS-positive tumors, lower MHC class II and co-stimulatory molecule expression on dendritic cells within TLS, and altered HEV/endothelial markers that hinder TLS maturation. In addition, PTEN loss and high mTORC1 activity will amplify PAK, YAP/TAZ, and TLR/NO signaling, further disrupting immune cell trafficking and APC function. Consequently, TLS-rich tumors with intact PTEN will show improved overall survival, whereas TLS-rich tumors with PTEN loss and high mTORC1 will not exhibit the TLS-associated survival benefit. In the TCGA-BRCA cohort, TLS-rich PTEN-intact tumors will have better OS than TLS-poor PTEN-intact tumors, while TLS-rich PTEN-deficient tumors with high mTORC1 activity will have OS similar to TLS-poor tumors. This defines a TLS dysfunction phenotype that could be targeted by PI3K/mTOR pathway modulation to restore TLS activity and anti-tumor immunity in a subset of breast cancers.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	PTEN, mTORC1
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PTEN: PTEN loss to altered chemokine-driven recruitment, endothelial/antigen-presentation changes, and a specific TIL phenotype tied to prognosis, providing a direct mechanistic route for TLS dysfunction.

mTORC1: mTORC1 path (PTEN → PI3K/AKT → mTORC1 negatively correlates with TLS-associated TIL phenotypes).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CXCL chemokines	Supporting modifier from knowledge graph
Secondary	TIL cluster C4	Supporting modifier from knowledge graph
Secondary	PhenoTIL clusters	Supporting modifier from knowledge graph
Secondary	Antigen-presentation machinery	Supporting modifier from knowledge graph
Secondary	Endothelial cells (ECs)	Supporting modifier from knowledge graph
Secondary	PAK pathway	Supporting modifier from knowledge graph
Secondary	YAP/TAZ	Supporting modifier from knowledge graph
Secondary	TLR/NO signaling	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 2: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CXCL chemokines: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TIL cluster C4: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.